

Total No. of Questions : 10]

SEAT No. :

P3628

[Total No. of Pages : 3

[5154]-673

B.E. (Computer Engineering)

SMART SYSTEM DESIGN & APPLICATION

(2012 Pattern) (End Semester)

Time : 2.30 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Questions Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data jf necessary
- 4) Figures to the right indicate marks.

**Q1)** a) Explain the concept of hierarchical planning along with example. [6]

b) Define artificial intelligence and the role of intelligent systems in smart applications. [6]

c) Using CSP, explore the search space to solve the following cryptarithmic problem [8]

LOGIC

+LOGIC

PROLOG

OR

**Q2)** a) Write short note on foundations of artificial intelligence. [6]

b) Explain CSP. Using CSP, explore the search space to solve the following crypt-arithmetic problem. TWO + TWO = FOUR. [6]

c) What is Knowledge Engineering? Explain the steps. [8]

P.T.O.

**Q3)** a) What is the basic inference task that must be solved in a generic temporal model? [6]

b) Write a short note on decision trees with suitable examples. [6]

OR

**Q4)** a) Write a short note on Kalman filters. [6]

b) What are axioms of probability? Explain how to derive the useful facts from basic axioms with suitable examples. [6]

**Q5)** a) Explain in brief the concept of Support Vector Machine and its applications. [6]

b) Write a note on Artificial Neural Network. [6]

OR

**Q6)** a) Write short note on : [6]

i) Active Learning.

ii) Reinforcement Learning

b) What is smart system? Write its applications. [6]

**Q7)** a) What are the Information Retrieval characteristics? How to evaluate and refine Information Retrieval System. [6]

b) Write a note on Robot Hardware. [6]

OR

**Q8)** a) What are the methods of handling uncertain knowledge? Write down the major challenges in handling uncertain knowledge. [6]

b) Explain Text Classification with suitable example. [6]

- Q9)** a) What is Machine Learning? What are the different paradigms? [6]  
b) What is Natural Language Processing(NLP)? Describe any 2 applications of NLP. [8]

OR

- Q10)** a) Explain Syntactic analysis (Parsing). [6]  
b) Write a short note on: [8]  
i) Machine Translation  
ii) Speech Recognition

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